Experimental design: studies, estimands, determinism, and power

The generative model of sec. 8.10, the learning operators of sec. 8.15,

Experimental design (part 2)

and the corruption process of sec. 8.19 are exercised by 9 studies, including the contaminated-sentinel robustness sweep (Study 4).

Experimental design (part 3)

The shared configuration (seed budget, colony size, contamination rates, divergences, trial counts, and the statistics settings) is read from `experiment:` in manuscript rendering configuration; the remaining per-study parameters are tested code defaults in `src/fedference/experiments/`.

Experimental design (part 4)

No value is hard-coded in the manuscript, and each token below resolves to the same configuration the code executed.

Determinism through fixed seeds and generated variables

All stochastic steps draw from explicitly seeded generators (`np.random.default_rng`); the global `np.random` state is never touched. The single-run studies use the first configured seed (0),

Determinism through fixed... (part 2)

and the across-seed studies enumerate the deterministic seed list $0, \ldots, n_{\text{seeds}} - 1$.

Determinism through fixed... (part 3)

Re-running with the same seed reproduces every number in the results bit-for-bit, so the bootstrap confidence intervals and paired-test p-values of sec. 8.27 are themselves deterministic functions of the seed.

Determinism through fixed... (part 4)

The figure layer follows the same provenance rule. Captions are written to be self-contained, with axes, resampling units, deterministic runs, truncated axes, and error-band status disclosed in the caption rather than left to inference (Rougier et al. 2014; Midway 2020).

Determinism through fixed... (part 5)

This is why several results figures state “single deterministic run” or “no error band” even when the inferential evidence appears in an adjacent table.

Study suite and contamination sweep

Studies 1–3 implement reduced categorical protocols that are source-mechanism analogues of the belief-sharing, language-acquisition, and model-reduction mechanisms discussed by Friston et al. (2024) on the sentinel POMDP; they are not exact source-protocol figure replications.

Study suite and... (part 2)

The sweep adds the FedGVI robustness contribution (Mildner et al. 2025).

Unless a study specifies otherwise, the global defaults are $n_{\text{seeds}} = 480$ independent seeds and $n_{\text{trials}} = 960$ matched trials per condition.

Study suite and... (part 3)

The inventories in tbls. 1, 2 separate each study's declared measure from its compact execution parameters so neither meaning is compressed at presentation scale.

Study suite and… (part 4)

Study	Declared measure
1 — Belief sharing	Communicating versus incommunicado colony free energy

Study suite and… (part 5)

Study	Key parameters
1 — Belief sharing	`n_agents = 7`; `acuity = 0.55`
2 — Language acquisition	`num_steps = 24`

Study suite and… (part 6)

Study 1 — belief sharing. A colony of 7 sentinels at the deliberately low acuity 0.55 each infer the creature location (eq. 16) and share beliefs through the log-linear pool (eq. 9). We compare a *communicating* colony against an *incommunicado* one of the same size and seed,

Study suite and... (part 7)

scoring each by the mean variational free energy of eq. 17.

Study suite and… (part 8)

Two protocol details: state inference in this study substitutes a flat (uniform) prior for D, and each belief's free energy is scored against the pooled evidence of all agents' observations (the disclosure carried in sec. 10.2). The across-seed sample is $n = 480$ seeds,

Study suite and... (part 9)

one colony-mean free energy per seed (fig. 12, fig. 13).

Study suite and... (part 10)

Study 2 — language acquisition. Each configured seed runs one sentinel trajectory using the conjugate Dirichlet update (eq. 18) over 24 steps. We record the KL descent of eq. 19 at each ordered step,

Study suite and... (part 11)

giving 25 points per trajectory and $n = 480$ independent seed trajectories for the pointwise interval in fig. 14.

Study suite and... (part 12)

Study 3 — configured BMR sign control. A redundant model reduction and a supported one are scored by the BMR free energy of eq. 20 over $n = 4$ candidate states (fig. 15). This fixed-posterior comparison tests the expected signs for two declared prunings;

Study suite and... (part 13)

it is not a structure-discovery experiment.

Study 4 — robustness sweep. The sweep varies two factors on a colony of 7 sentinels of which 2 are saboteurs (`confident_wrong`, sec. 8.20):

Contamination rate

The contamination rate spans $\{0, 0.225, 0.45, 0.675, 0.9\}$ and is the convex-mix weight of eq. 21 toward the confident-wrong delta.

Server robustness setting

The server robustness setting is named with FedGVI client-loss/divergence vocabulary for cross-reference only: $\{KLD, RKL, AR, beta, rcce\}$.

Server robustness setting (part 2)

`KLD` is the non-robust Friston / standard-Bayes baseline (server robustness 0, eq. 10) and serves as the design's negative control. The recovery identity guarantees that it reproduces the naive log-linear pool exactly,

Server robustness setting (part 3)

so every contrast uses the proved un-robustified server rather than a separately tuned competitor.

Server robustness setting (part 4)

The remaining labels $\{RKL, AR, beta, rcce\}$ each select a fixed `robust_aggregate` down-weighting constant. The executed mapping is KLD (c=0.00), RKL (c=1.50), AR (c=1.30), beta (c=1.70), rcce (c=1.60), defined in `fedference.experiments._common`.

Server robustness setting (part 5)

None of these labels invoke the client-side `generalized_posterior` update of sec. 8.3 or the divergence family of sec. 8.6. This sweep exercises only the server-side heuristic axis of sec. 6.3 (`robust_aggregate`),

Server robustness setting (part 6)

never the client-side source-conditional theorem lane.

The executed per-divergence down-weighting strengths are fixed constants defined in `fedference.experiments._common` and recorded in the run reports.

Server robustness setting (part 7)

The headline verdict pairs 960 independent trials at the fixed contamination rate 0.800 — heavy contamination that degrades the naive pool while staying below the pure-veto cliff (fig. 16, fig. 18). Each trial contributes one matched (naive, robust) accuracy pair,

Server robustness setting (part 8)

so the replication unit is the trial and the estimand is the within-trial accuracy difference at that single rate.

Server robustness setting (part 9)

A complementary exploratory point-estimate logistic-regression proxy compares joint NLL/L2 and RCCE/L2 configurations under flipped-label contamination (fig. 21). Because both loss and shrinkage change,

Server robustness setting (part 10)

this is a composite loss-and-L2 contrast rather than an RCCE-only effect; it represents no weight-space posterior and does not implement a FedGVI generalized-posterior objective.

Server robustness setting (part 11)

Study 5 — disjoint-FOV moving world. This extension (sec. 17.7) places 3 agents on a 2-slot disjoint-FOV track to test the necessity of communication when agents cannot observe the same positions.

Server robustness setting (part 12)

Isolated agents are compared against EFE-guided communicating agents on accuracy across the moving sentinel's trajectory.

Server robustness setting (part 13)

Five structural extension studies (Studies 5–9, Supplementary sections) build on the same POMDP substrate and are described there: the moving disjoint-FOV sentinel (sec. 17.7), the 2-level hierarchical POMDP (sec. 17.9),

Server robustness setting (part 14)

the N-level extension (sec. 17.11), the 2-D sensitivity sweep (sec. 17.13), and parameter recovery (sec. 18).

Sample size and prospective statistical power

The verdict design answers a deliberate question: pair *many* trials at *one* high contamination rate rather than spread few trials across the rate curve.

Sample size and... (part 2)

A matched-pairs Wilcoxon test gains power from the number of matched pairs, so concentrating $n = 960$ paired trials at the single rate 0.800 gives the test the resolution to detect the robustness effect;

Sample size and... (part 3)

scattering the same budget across 5 rates would dilute every contrast.

The across-seed studies are powered separately, with $n = 480$ seeds for belief sharing and $n = 480$ independent seed trajectories for language acquisition.

Sample size and... (part 4)

The 25 ordered learning points are repeated measures within each seed, not additional independent replicates, so the language interval does not count time points as samples.

Sample size and... (part 5)

The structural-extension and cross-study summary tier uses $n = 128$ independent seeds and $n = 40$ matched trials per contamination rate for its robustness row. Trials and clients are nested within a seed and are reduced before across-seed inference; they are not additional independent replicates.

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Sample size and... (part 6)

The bounded red-team review grid is a separate source-bound analysis profile. It uses 160 deterministic seed replicates, with 24 trials nested within each seed and scenario/rate cell, and retains the registered rates 0, 0.2, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9 across the finite attack union clean, confident wrong, permutation, byzantine,

Sample size and... (part 7)

drift, label noise, uniform.

Sample size and… (part 8)

Its independent unit is the configured seed within a declared cell; cells that share design structure are not treated as independent worlds. Robust operating points, method settings, and rate profiles are fixed before the review run.

Sample size and... (part 9)

The grid reports selection-free contrasts and keeps clean, uniform, label-noise, permutation, confident-wrong, Byzantine, and drift controls visible. Completion of this finite review does not close server-heuristic characterization, leakage-free calibration, external-data, or protocol-reconstruction phases.

Sample size and... (part 10)

We do not merely assert adequacy — we compute the design power implied by the observed effect.

Sample size and... (part 11)

For the headline robust method (RKL) the observed-effect design power of the paired Wilcoxon at the run's $n = 960$, computed at $\alpha = 0.05$ against the directional alternative `greater` (robust accuracy exceeds naive), is 1.0000.

Sample size and... (part 12)

The power approximation uses the deterministic noncentral-normal approximation of sec. 8.27, deflated by the Wilcoxon's Pitman asymptotic relative efficiency, so it is approximately calibrated for the signed-rank test actually run (exact under a normal-shift alternative;

Sample size and... (part 13)

the power computation is one-sided while the reported p-values are two-sided).

Sample size and... (part 14)

To bound a confirmatory replication, the prospective sample size needed to reach the target power 0.80 at the observed effect is $n = 5$ matched trials — the explicit sample-size budget a follow-up study should adopt.

Sample size and... (part 15)

These power quantities characterize the *server-side* aggregation contrast; per the honesty contract of sec. 6.3 they do not certify the per-client β/rcce guarantees, which are pinned by the locked core (sec. 9) rather than by these aggregation-level statistics.

Software environment and configuration fingerprint

The FedGVI core, the POMDP studies, and the logistic-regression baseline are pure NumPy / SciPy — no GPU and no network.

Software environment and… (part 2)

The deterministic-MLP neural complement (sec. 10.6) additionally uses PyTorch (CPU);

the analysis pipeline executes it and emits its numbers as tokens exactly like every other result when the `torch` optional extra is installed,

Software environment and... (part 3)

and otherwise records a skipped status with unavailable-value sentinels instead of silently fabricating neural results.

Software environment and... (part 4)

Versions and platform — including the PyTorch version used for the MLP complement — are recorded automatically in sec. 13.